

WAKAMATSU, Japan

WMO: 475700

Lat: 37.4884N

Lon: 139.9104E

Elev: 701

StdP: 14.33

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.6	23.1	15.7	12.5	25.8	17.7	13.8	27.7	18.3	30.9	16.5	31.1	3.1	300	0.405

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	16.6	92.3	76.3	89.7	75.5	86.8	74.1	77.7	88.6	76.5	86.5	75.5	84.3	7.6	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.7	133.4	81.9	73.8	129.2	81.1	72.9	125.1	80.3	41.7	88.8	40.6	87.0	39.6	84.4	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.4	14.4	12.6	DB	14.4	95.9	3.5	2.0	11.9	97.3	9.8	98.5	7.9	99.6	5.3	101.1
			WB	13.9	79.7	3.5	0.9	11.4	80.3	9.3	80.8	7.4	81.3	4.8	82.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.3	31.2	32.3	39.0	50.7	61.8	69.5	76.4	78.4	70.3	58.4	45.8	36.2
	DBStd	17.37	4.03	4.38	5.45	6.73	5.55	4.44	4.54	4.50	5.88	5.77	5.60	4.81
	HDD50	2079	583	496	343	71	2	0	0	0	0	5	150	429
	HDD65	5043	1048	916	806	429	127	12	0	0	18	217	577	893
	CDD50	3645	0	0	2	92	368	585	820	881	608	265	24	0
	CDD65	1138	0	0	0	1	29	148	355	416	176	13	0	0
	CDH74	8563	0	0	0	31	324	844	2650	3617	1049	48	0	0
	CDH80	2985	0	0	0	3	57	184	969	1472	293	7	0	0
Wind	WSAvg	4.5	5.1	5.2	5.3	5.4	5.1	4.5	3.8	3.9	3.9	3.8	3.9	4.6
Precipitation	PrecAvg	52.6	3.3	2.5	3.0	3.1	3.5	4.9	7.8	6.5	6.2	4.8	3.3	3.8
	PrecMax	65.2	8.0	3.9	5.6	5.5	6.5	8.6	15.3	20.4	11.2	13.7	6.3	8.0
	PrecMin	39.2	1.6	0.6	0.9	0.4	1.5	2.5	2.7	1.5	0.4	1.3	1.1	1.0
	PrecStd	7.0	1.5	0.8	1.1	1.4	1.4	1.7	2.8	3.9	2.4	3.1	1.3	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.9	49.4	65.0	78.4	85.3	88.1	94.4	96.3	90.8	79.4	65.1	54.2
		MCWB	41.1	42.3	50.7	58.5	64.1	69.6	77.1	77.3	74.7	68.1	56.5	49.5
	2%	DB	41.4	45.1	57.9	72.5	81.2	84.7	91.7	93.3	87.0	74.2	60.8	49.1
		MCWB	37.6	39.3	46.6	55.3	62.1	68.2	76.4	76.4	74.0	64.1	54.1	45.1
	5%	DB	39.1	42.0	52.9	68.2	77.6	81.9	88.9	90.9	83.3	71.1	58.1	46.2
		MCWB	35.6	37.4	44.5	53.7	60.8	67.1	75.3	76.2	72.0	62.5	52.6	42.6
	10%	DB	37.1	39.4	48.9	63.9	74.1	79.0	85.6	88.0	80.0	68.1	55.2	43.6
		MCWB	34.2	35.5	42.8	51.8	59.7	66.4	74.2	75.4	70.5	60.5	51.0	40.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.3	43.6	52.0	63.0	67.9	73.7	78.9	79.5	76.7	69.6	58.6	50.5
		MCDB	44.5	48.1	60.9	69.3	76.8	81.0	90.6	91.5	86.9	77.3	61.7	53.2
	2%	WB	38.3	40.3	48.6	58.2	65.2	71.7	77.4	78.1	74.8	66.4	55.8	45.8
		MCDB	40.7	43.8	55.3	68.0	75.0	78.2	88.2	89.3	84.0	71.7	59.4	48.3
	5%	WB	36.1	37.8	45.8	55.7	63.4	70.1	76.3	77.1	73.4	64.1	53.7	43.0
		MCDB	38.3	41.3	51.3	64.7	72.7	77.0	85.9	87.6	81.2	68.9	56.8	45.4
	10%	WB	34.5	35.8	43.0	53.6	61.7	68.7	75.3	76.1	71.9	62.0	51.3	40.9
		MCDB	36.6	39.0	48.3	61.5	70.7	75.8	83.4	85.5	78.2	66.9	54.7	43.2
Mean Daily Temperature Range	5% DB	MDBR	9.9	12.0	15.8	20.4	20.6	16.6	15.1	16.6	15.9	16.4	14.3	10.3
		MCDBR	12.0	16.2	25.4	29.1	28.2	22.4	20.3	20.9	19.7	19.6	18.8	14.4
		MCWBR	8.6	11.2	15.3	14.1	11.8	8.6	7.5	7.1	8.3	10.0	11.9	10.6
	5% WB	MCDBR	11.4	15.5	22.8	23.6	21.1	16.3	17.8	18.7	17.7	15.9	15.8	13.6
		MCWBR	8.9	11.3	15.0	12.8	10.4	7.2	7.2	7.1	7.9	8.9	11.4	10.7
Clear-Sky Solar Irradiance	taub	0.403	0.453	0.533	0.583	0.598	0.617	0.621	0.613	0.526	0.470	0.426	0.403	
	taud	1.958	1.831	1.749	1.717	1.727	1.747	1.797	1.811	1.986	2.045	2.078	2.048	
	Ebn at Noon	232	236	230	228	228	223	221	219	231	230	225	223	
	Edh at Noon	43	56	67	73	74	72	68	66	52	45	38	37	
All-Sky Solar Radiation	RadAvg	572	828	1142	1413	1629	1539	1422	1454	1185	918	650	506	
	RadStd	48	68	87	136	167	103	219	164	99	83	45	44	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page